Ahmad Ghasemi

+1 (906) 231-5803 | aghasemi@umass.edu | https://ahdghasemi.github.io | LinkedIn | Google Scholar
University of Massachusetts, Amherst, MA | Work authorization: U.S. Permanent Resident

Research Profile

I am a graph machine learning researcher developing theoretically grounded methods for structured decision-making problems. My work studies when local graph representations and message-passing architectures can generalize across problem instances, and how these insights can support reliable ML-guided optimization, online decision-making, and learning under uncertainty.

Core research areas: graph neural networks; statistical learning theory; sample complexity; locality and message passing; structure-aware generalization; learning-augmented algorithms; optimization-aware learning; adversarial robustness; uncertainty-aware and efficient inference.

Research leadership: Co-PI and key personnel on externally funded and submitted NSF projects involving graph-based modeling, trustworthy neural learning, multi-agent decision systems, and learning-and-control evaluation pipelines.

Research Funding and Proposal Leadership

Active — **NSF CPS Small, Co-PI**, AERIAL: AI-Embedded Responsive Intelligent Agents with Trajectory-Induced Digital Twin Learning, CNS-2528914, $599,961, 08/2025–07/2028. *PI:* H. Pishro-Nik. *Role:* Lead investigator for graph-based modeling, learning-and-control, budget-aware neural inference, robustness evaluation, and performance metrics for multi-agent networked decision systems.

Under Review — **NSF CyberAI SFS, Senior Personnel**, CyberAI Innovation: An Educational Security Testbed for Networked AI Agents, $499,680. *PI:* H. Pishro-Nik, *Co-PI:* A. Houmansadr. *Submitted:* Apr. 2026. *Role:* Contributed to the technical vision, AI-agent testbed design, uncertainty-aware reasoning activities, and research-to-education integration.

Research Program and Selected Contributions

- **Graph neural network theory and topology-dependent generalization.** Established minimax lower bounds and structural regimes for message-passing graph neural networks, clarifying when learnability is governed by graph topology, label availability, feature dimension, and spectral–homophily structure. This work connects statistical learning theory with the design and evaluation of graph-based learning systems.
- **Learning-augmented decision systems under uncertainty.** Developed graph learning and neural inference methods for decision-making in online, networked, and resource-constrained systems, where models must operate under uncertainty, limited information, latency constraints, and changing system conditions. This work includes graph-based resource allocation, UAV/network coordination, and evaluation pipelines for learned decision policies.
- **Robust graph learning and adversarial evaluation.** Developed adversarial evaluation and robustness methods for graph-based resource allocation, with applications to wireless and cyber-physical decision systems. This work identifies failure modes of learned graph policies under adversarial perturbations, imperfect system information, and distribution shift.
- **Resource-aware and constrained graph learning.** Designed compact, low-rank, and constrained neural models for inference under memory, latency, energy, and communication limits. This includes tiny graph neural networks for radio resource management and constrained low-rank training methods for graph neural networks.
- **Efficient reasoning and confidence-guided inference.** Contributed to work on confidence-guided early stopping for efficient inference, connecting uncertainty-aware model behavior with computational efficiency in large-model reasoning systems.

Professional Appointments

Postdoctoral Research Fellow, University of Massachusetts, Amherst, MA — **05/2024 - Present**
Supervisor: Hossein Pishro-Nik.
Focus: Graph neural networks and structured learning for decision-making systems, including GNN sample complexity, topology-dependent generalization, robustness under distribution shift/adversarial perturbations, and efficient inference.

Adjunct Faculty (Instructor of Record / Co-Instructor), University of Massachusetts, Amherst, MA — **09/2023 - Present**
Courses: Machine Learning for Engineers, Foundations of Generative Models, Digital Signal Processing, and Digital Image Processing (graduate/undergraduate; project-based delivery).

Education

Ph.D., Data Science, Worcester Polytechnic Institute (WPI), Worcester, MA — **2023**
Dissertation: Intelligent and Resilient Resource Allocation in mmWave Wireless Communications

Research areas: graph neural networks, graph-based optimization, learning-based resource allocation, adversarial robustness, and decision-making under uncertainty.

Advisor: Reza Zekavat

M.Sc., Electrical and Computer Engineering, Shiraz University, Shiraz **2012**

Dissertation: Machine Learning-Based Spectrum Allocation in Cognitive Radio Networks

Advisors: Mohammad Ali Masnadi-Shirazi; Mehrzad Biguesh

B.S., Electrical and Computer Engineering, University of Sistan and Baluchestan, Zahedan **2007**

Publications

* indicates co-first authorship.

Peer-Reviewed Conference Proceedings

[C1] **A. Ghasemi** and H. Pishro-Nik, “Minimax Sample Complexity of Graph Neural Networks: Lower Bounds and Structural Effects,” The Fourteenth International Conference on Learning Representations (ICLR) 2026. OpenReview PDF.

[C2] M. Malekzadeh, **A. Ghasemi**, and H. Pishro-Nik, ”Polar-Grid Graph Attention for Finite-Horizon Density-Aware UAV Coverage,” IEEE Vehicular Technology Conference, Boston, MA, 2026.

[C3] M. Malekzadeh*, **A. Ghasemi***, and H. Pishro-Nik, “Constant-Speed Trajectory Processes for Uniform Coverage in UAV Networks,” 59th Annual Conference on Information Science & Systems (CISS), Baltimore, Maryland, USA, 2025.

[C4] **A. Ghasemi** and H. Pishro-Nik, “Tiny Graph Neural Networks for Radio Resource Management,” in *Proceedings of tinyML Research Symposium (tinyML Research Symposium’24)*, ACM, Burlingame, CA, USA, 2024, pp. 1–7.

[C5] **A. Ghasemi**, M. Moradikia, R. Zekavat and H. Pishro-Nik, “Adversarial Attacks Targeting Point-to-Point Wireless Networks,” 2024 IEEE 99th Vehicular Technology Conference (VTC2024-Spring), Singapore, Singapore, 2024, pp. 1–6.

[C6] **A. Ghasemi**, E. Zeraatkar, M. Moradikia, and R. Zekavat, “Adversarial Attacks on Resource Management in P2P Wireless Communications,” 2023 IEEE International Conference on Wireless for Space and Extreme Environments (WiSEE), Aveiro, Portugal, 2023, pp. 148–153.

[C7] **A. Ghasemi** and R. Zekavat, “On Eigenvalue Distribution of Imperfect CSI in mmWave Communications,” 2022 IEEE USNC-URSI Radio Science Meeting (Joint with AP-S Symposium), Denver, CO, USA, 2022, pp. 56–57. (***Travel Award***)

[C8] **A. Ghasemi** and R. Zekavat, “Joint Hybrid Beamforming and Dynamic Antenna Clustering for Massive MIMO,” 2020 29th Wireless and Optical Communications Conference (WOCC), Newark, NJ, USA, 2020, pp. 1–6. (***Best Paper Award***)

[C9] **A. Ghasemi**, A. F. Jahromi, M. A. Masnadi-Shirazi, M. Biguesh and F. Ghasemi, “Spectrum Allocation Based on Artificial Bee Colony in Cognitive Radio Networks,” 6th International Symposium on Telecommunications (IST), Tehran, Iran, 2012, pp. 182–187.

[C10] **A. Ghasemi**, A. F. Jahromi, M. A. Masnadi-Shirazi, M. Biguesh and F. Ghasemi, “Spectrum Allocation with Control of Interference Based on Differential Evolution Algorithm Between Cognitive Radio Users,” Proc. 20th Iranian Conf. Elect. Eng. (ICEE), 2012.

[C11] Z. Iqbal, S. Nooshabadi, K. Jadi and **A. Ghasemi**, “Sensor Cooperation and Decision Fusion to Improve Detection in Cognitive Radio Spectrum Sensing,” 9th IEEE Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON), New York, NY, USA, 2018, pp. 276–281.

Peer-Reviewed Journal Articles

[J1] M. Malekzadeh*, **A. Ghasemi***, and H. Pishro-Nik, ”Robust UAV Trajectory Design for Non-Uniform Coverage,” IEEE Communications Letters, vol. 30, pp. 188-192, 2026.

[J2] **A. Ghasemi**, E. Zeraatkar, M. Moradikia, and R. Zekavat, “Adversarial Attacks on Graph Neural Networks-Based Spatial Resource Management in P2P Wireless Communications,” *IEEE Transactions on Vehicular Technology*, vol. 73, no. 6, pp. 8847–8863, June 2024.

[J3] **A. Ghasemi** and S. A. Zekavat, “Low-Cost mmWave MIMO Multi-Streaming via Bi-Clustering, Graph Coloring, and Hybrid Beamforming,” *IEEE Transactions on Wireless Communications*, vol. 20, no. 7, pp. 4113–4127, July 2021.

[J4] **A. Ghasemi**, M. Masnadi-Shirazi, M. Biguesh, and F. Qassemi, “Channel Assignment Based on Bee Algorithms in Multi-hop Cognitive Radio Networks,” *IET Communications*, 8(13), 2356–2365, 2014.

[J5] R. Zekavat, R. M. Buehrer, G. D. Durgin, L. Lovisolo, Z. Wang, T. Goh, and **A. Ghasemi**, “An Overview on Position Location: Past, Present, Future,” *International Journal of Wireless Information Networks*, 28, 45–76 (2021).

Peer-Reviewed Workshop Papers

[W1] E. Aghazade, **A. Ghasemi**, H. Beyhaghi, and H. Pishro-Nik, “CGES: Confidence-Guided Early Stopping for Efficient Inference,” Efficient Reasoning Workshop, NeurIPS 2025. ArXiv.

Preprints and Manuscripts Under Review

[P1] S. Amini*, **A. Ghasemi***, and H. Pishro-Nik, "Constrained Optimization for Low-Rank Training of Graph Neural Networks," Under review (*IEEE Transactions on Signal Processing*). *Preprint available upon request.*

[P2] E. Aghazade, M. Malekzadeh, **A. Ghasemi**, and H. Pishro-Nik, "Ergodic Trajectory Design by Learned Pushforward Maps: Provable Coverage via Conditional Flow Matching," Under review (*NeurIPS 2026*). ArXiv.

[P3] **A. Ghasemi**, M. Malekzadeh, E. Aghazade, and H. Pishro-Nik, "Tiny Flow Matching: Constrained Low-Rank OT-CFM for Deployable Ergodic UAV Coverage," Under review (*IEEE International Workshop on Machine Learning for Signal Processing*). *Preprint available upon request.*

Book

[B1] F. Ghasemi, **A. Ghasemi**, "Numerical Methods: Basics and coding in MATLAB," Zolal-e-Sabz Press, 2020.

Research Mentorship

Research mentor / co-mentor: 3 Ph.D. students, 1 M.Sc. student, and 5 undergraduate researchers at UMass Amherst.

Mentoring areas: graph neural networks, UAV trajectory learning, efficient LLM inference, neural architecture search, generative AI, and AI-tutor systems.

Selected outcomes: IEEE journal and conference publications; NeurIPS workshop acceptance; EMNLP/NeurIPS/major-venue submissions; honors thesis direction; reproducible evaluation harnesses.

Spring 2025 – Present — **Ehsan Aghazadeh**, Ph.D. student, Computer Science — UMass Amherst
Focus: Efficient and accurate LLM test-time scaling.
Outcome: NeurIPS workshop acceptance and EMNLP submission.

2024–Present — **Masoud Malekzadeh**, Ph.D. student, Electrical and Computer Engineering — UMass Amherst
Focus: Graph-based and learning-based unmanned aerial vehicle (UAV) trajectory design.
Outcome: IEEE journal and conference publications.

Fall 2025 – Present — **Giap Hoang Nguyen**, UG student, Mathematics and Computer Science — UMass Amherst
Focus: UAV orchestration using graph neural networks.
Outcome: Project completed; manuscript/report in preparation.

Fall 2025 — **Thuyen Pham**, UG student, Mathematics and Computer Science — UMass Amherst
Focus: Automated imaging for butterfly identification.
Outcome: Research direction adopted as Honors thesis.

Summer 2025 — **Huy Gia Cao, Jake Reid, Mykaala Firdaus, Ritik Shah**, undergraduate researchers, Electrical and Computer Engineering — UMass Amherst
Focus: LLM fine-tuning and AI-tutor prototyping.
Outcome: Fine-tuned LLM; curated training dataset; completed AI-Tutor prototype.

Talks

March 2026 — **University of Texas, Tyler** — Invited
Trustworthy and Resource-Aware AI: From Graph Learning Theory to Deployment

March 2026 — **Florida Polytechnic University, Lakeland** — Invited
Building Reliable and Efficient ML for Real-World Decision-Making

March 2026 — **California State University, Long Beach** — Invited
Trustworthy and Secure Edge AI under Real-World Constraints

May 2024 — **TinyML Research Symposium** — Burlingame, CA
Tiny Graph Neural Networks for Radio Resource Management

May 2023 — **NEWSDR: New England Workshop on Software-Defined Radio** — Worcester, MA
Adversarial Attacks on Graph Neural Network-based Wireless Communications

Professional Service

- NSF CISE and TTP-T Panels Reviewer, 2026.
- Reviewer: *ICLR*; *IEEE Transactions on Wireless Communications*; *IEEE Transactions on Vehicular Technology*; *IEEE Transactions on Communications*; *IEEE Transactions on Information Forensics and Security*; EURASIP Journal on Wireless Communications and Networking.
- Organizing Chair, IEEE WiSEE 2020.
- Curriculum Committee Member, UMass iCons track "AI and the Future of Work," 2024–present.

Teaching Experience

Teaching areas: machine learning, generative models, probability/statistics, data science foundations, graph neural networks, signal processing, image processing, and trustworthy/efficient AI.

Teaching effectiveness: Recent UMass Amherst instructor effectiveness **4.7/5.0**; well prepared **5.0/5.0**; used class time well **5.0/5.0**; showed interest in helping students learn **5.0/5.0**.

Instructor of record / Co-Instructor

Fall 2025 — **ECE690: Foundations of Generative Models (Graduate)** — UMass Amherst
Co-Instructor

Fall 2025 — **ECE150: Better Decisions by Human & AI (Undergraduate)** — UMass Amherst
Instructor

Spring 2025, Summer 2024, Spring 2024 — **ECE601: Machine Learning for Engineers (Graduate)** — UMass Amherst
Instructor / Co-Instructor

Fall 2024 — **ECE565: Digital Signal Processing & Representation (Graduate)** — UMass Amherst
Instructor

Summer 2024 — **ECE579: Math Tools for Data Science & Machine Learning (Graduate)** — UMass Amherst
Instructor

Fall 2023 — **ECE566: Digital Image Processing (Graduate)** — UMass Amherst
Instructor

Teaching Assistant

Spring 2022, Fall 2021, Fall 2020 — **DS517: Math Foundations for Data Science (Graduate)** — WPI
Graduate Teaching Assistant

Spring 2021 — **DS541: Deep Learning (Graduate)** — WPI
Graduate Teaching Assistant

Spring 2020 — **DS2010: Data Science II: Modeling & Data Analysis (Undergraduate)** — WPI
Graduate Teaching Assistant

Fall 2019 — **ECE2312: Signal and System Analysis (Undergraduate)** — WPI
Graduate Teaching Assistant

Course & Curriculum Development

2024–Present — **Curriculum Development for New iCons Track** — UMass Amherst
AI and the Future of Work, Curriculum Committee Member

Fall 2025 — **Developed New Graduate Course** — UMass Amherst
ECE690: Foundations of Generative Models

Spring 2024 — **Developed New Graduate Course** — UMass Amherst
ECE601: Machine Learning for Engineers

Industry and Applied Research Experience

Industry-Sponsored Research Intern, Ford Motor Company / Wireless Positioning Lab, Michigan Tech — **Summer 2019**

- Built reproducible perception and positioning experiments for autonomous-driving scenarios under sensing uncertainty and environmental perturbations.

Data Scientist, Hormozgan Electricity Distribution Co., Bandar Abbas — **2015–2018**

- Developed forecasting, anomaly detection, and decision-support workflows for power-distribution reliability.

Honors and Awards

Charles Kao Best Paper Award, 29th Wireless and Optical Communications Conference, Newark, NJ, USA — **2020**
Travel Award, School of Arts & Sciences, Worcester Polytechnic Institute, Worcester, MA, USA — **2022**
Nominee, TA of the Year Award, Worcester Polytechnic Institute, Worcester, MA, USA — **2022**